
MSML 605

Data Centric Development

Model Centric Development

- Build a model
 - Take the data and make the model fit that data.
 - Model centric - fix the data and improve the model.
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Data Centric Development

- Can you make your data better?
- Data augmentation.

Medical Imaging application (MRI or CT)

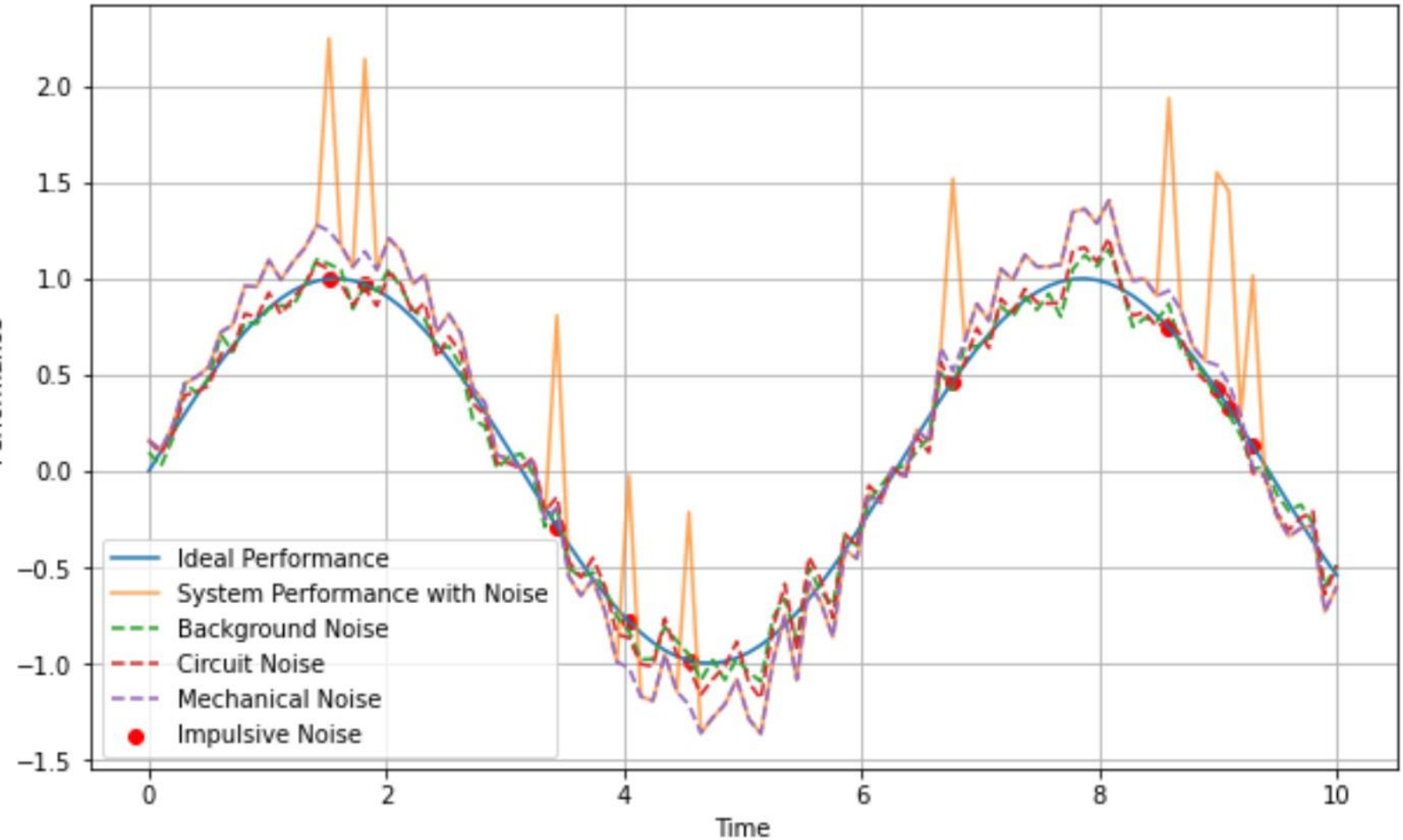
- Background sounds, for example, machinery noise or ambient clutter in the hospital
- Vibrational noise by nearby equipment or structural elements
- Scanner noise generated by the movement of mechanical components
- Electromagnetic interference from other electronic devices
- Thermal noise due to fluctuations in the electronic components
- Motion artifacts resulting from patient movement

Medical Imaging application (MRI or CT)

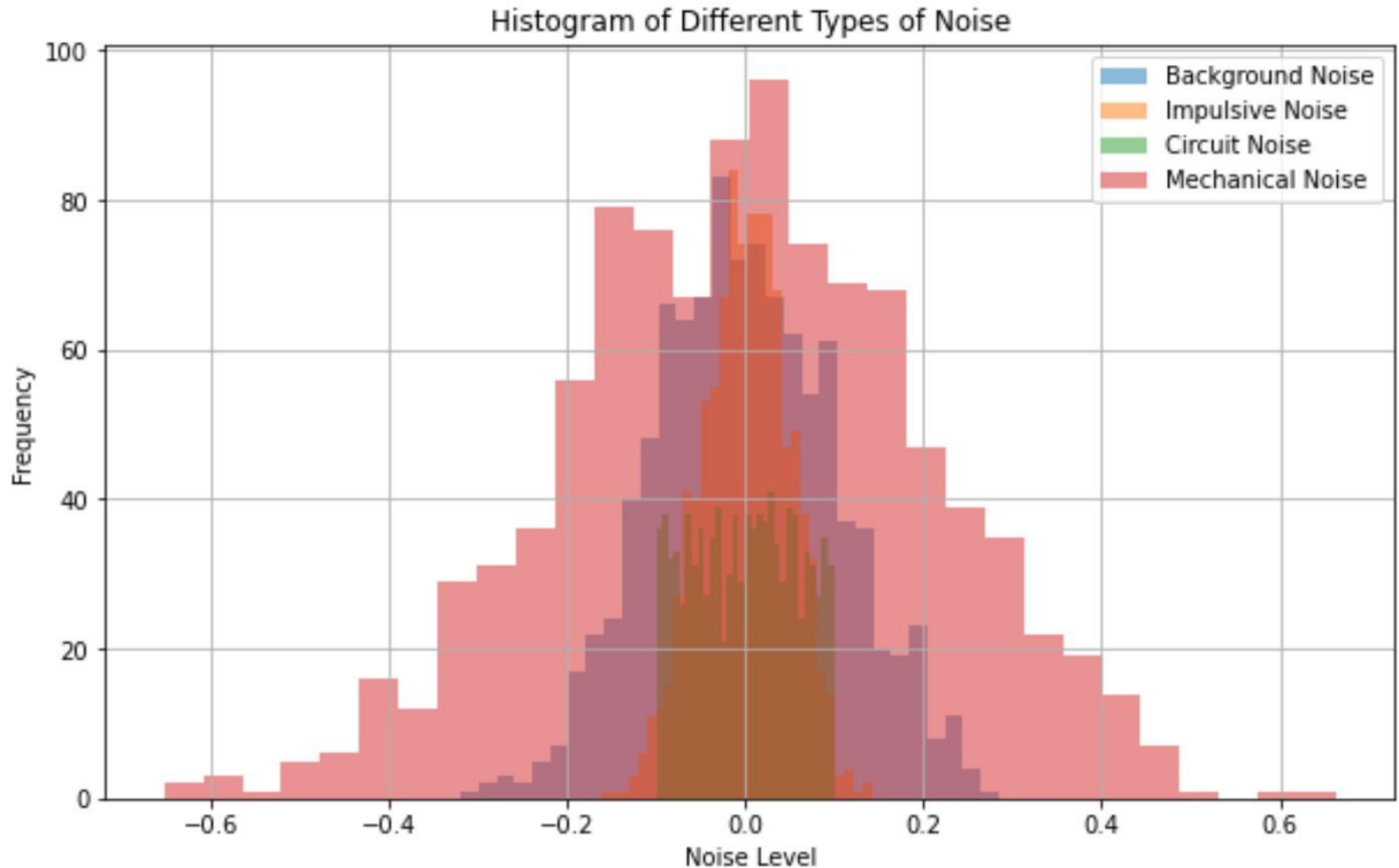
- External Noise
 - Environmental noise
 - Vibrational noise
 - System Noise
 - Scanner operation noise
 - Electromagnetic interference
 - Signal Noise
 - Thermal noise
 - Motion artifacts
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Medical Imaging application (MRI or CT)

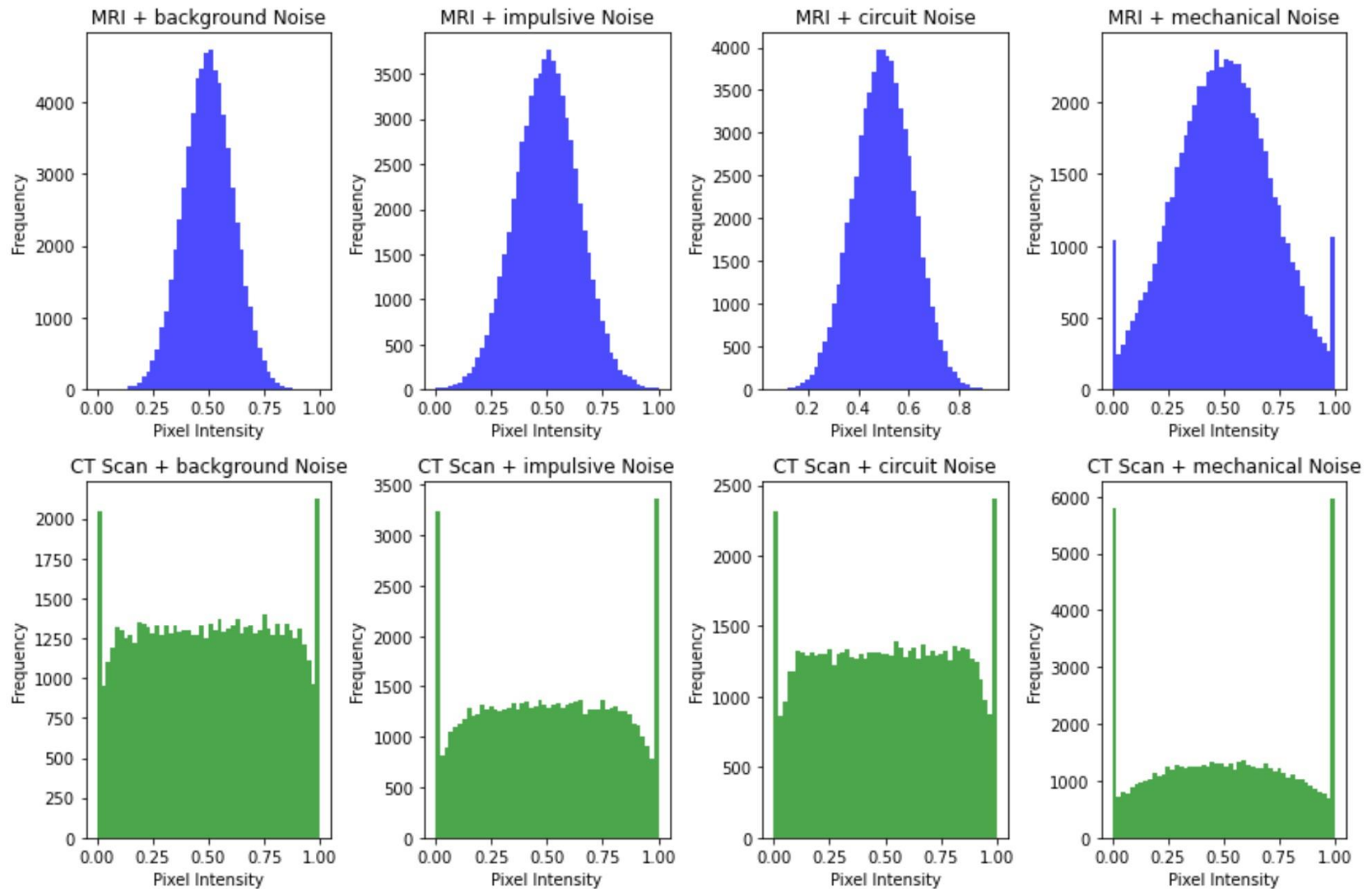
Impact of Different Types of Noise on System Performance



Medical Imaging application (MRI or CT)



Medical Imaging application (MRI or CT)



Data Augmentation

- Create examples to augment data.
 - Specifically for the examples where model does not do well.
 - Simple methods of creating new data
 - GANs maybe an overkill
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Data Iteration

- Repeat (hold model fixed)
 - Add data
 - Train a model
 - Error analysis

Content filtering

- Instead of collaborative filtering focus is now shifting to content filtering
 - Learning about the user and the product features
 - Cold start problem can be avoided unlike in collaborative filtering.
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Track Experiments

- Code versioning
 - Dataset tracking
 - Hyperparameters
 - Results - metrics
 - Save model.
 - Text files, spreadsheets, experiment tracking system
 - Using tools: Weights & biases, Comet, MLFlow, SageMaker Studio etc.
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